

Problem Statement

Business Context

The healthcare industry is rapidly evolving, with professionals facing increasing challenges in managing vast volumes of medical data while delivering accurate and timely diagnoses. The need for quick access to comprehensive, reliable, and up-to-date medical knowledge is critical for improving patient outcomes and ensuring informed decision-making in a fast-paced environment.

Healthcare professionals often encounter information overload, struggling to sift through extensive research and data to create accurate diagnoses and treatment plans. This challenge is amplified by the need for efficiency, particularly in emergencies, where time-sensitive decisions are vital. Furthermore, access to trusted, current medical information from renowned manuals and research papers is essential for maintaining high standards of care.

To address these challenges, healthcare centers can focus on integrating systems that streamline access to medical knowledge, provide tools to support quick decision-making, and enhance efficiency. Leveraging centralized knowledge platforms and ensuring healthcare providers have continuous access to reliable resources can significantly improve patient care and operational effectiveness.

Common Questions to Answer

- 1. Diagnostic Assistance:** "What are the common symptoms and treatments for pulmonary embolism?"
- 2. Drug Information:** "Can you provide the trade names of medications used for treating hypertension?"
- 3. Treatment Plans:** "What are the first-line options and alternatives for managing rheumatoid arthritis?"
- 4. Specialty Knowledge:** "What are the diagnostic steps for suspected endocrine disorders?"
- 5. Critical Care Protocols:** "What is the protocol for managing sepsis in a critical care unit?"

Objective

As an AI specialist, your task is to develop a RAG-based AI solution using renowned medical manuals to address healthcare challenges. The objective is to **understand** issues like information overload, **apply** AI techniques to streamline decision-making, **analyze** its impact on diagnostics and patient outcomes, **evaluate** its potential to standardize care practices, and **create** a functional prototype demonstrating its feasibility and effectiveness.

Data Description

The **Merck Manuals** are medical references published by the American pharmaceutical company Merck & Co., that cover a wide range of medical topics, including disorders, tests, diagnoses, and drugs. The manuals have been published since 1899, when Merck & Co. was still a subsidiary of the German company Merck.

The manual is provided as a PDF with over 4,000 pages divided into 23 sections.

Installing and Importing Necessary Libraries and Dependencies

```
# Installation for GPU llama-cpp-python
# uncomment and run the following code in case GPU is being used
!CMAKE_ARGS="-DLLAMA_CUBLAS=on" FORCE_CMAKE=1 pip install llama-cpp-python==0.2.28

# Installation for CPU llama-cpp-python
# uncomment and run the following code in case GPU is not being used
# !CMAKE_ARGS="-DLLAMA_CUBLAS=off" FORCE_CMAKE=1 pip install llama-cpp-python==0.2.
```

```
_____ 0.0/9.4 MB ? eta -:--:--
_____ 3.4/9.4 MB 101.2 MB/s eta 0:00:01
_____ 9.4/9.4 MB 208.2 MB/s eta 0:00:00
```

```
Installing build dependencies ... done
Getting requirements to build wheel ... done
Installing backend dependencies ... done
Preparing metadata (pyproject.toml) ... done
```

```
_____ 45.5/45.5 kB 583.3 MB/s eta 0:00:00
_____ 16.6/16.6 MB 392.9 MB/s eta 0:00:00
_____ 44.6/44.6 kB 599.7 MB/s eta 0:00:00
```

```
Building wheel for llama-cpp-python (pyproject.toml) ... done
```

```
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
```

```
tensorflow 2.19.0 requires numpy<2.2.0,>=1.26.0, but you have numpy 2.4.3 which is incompatible.
```

```
numba 0.60.0 requires numpy<2.1,>=1.22, but you have numpy 2.4.3 which is incompatible.
```

Note:

- After running the above cell, kindly restart the runtime (for Google Colab) or notebook kernel (for Jupyter Notebook), and run all cells sequentially from the next cell.
- On executing the above line of code, you might see a warning regarding package dependencies. This error message can be ignored as the above code ensures that all necessary libraries and their dependencies are maintained to successfully execute the code in **this notebook**.

```
!pip install huggingface_hub==0.35.3 pandas==2.2.2 tiktoken==0.12.0 pymupdf==1.26.5
```

```

_____ 60.9/60.9 kB 5.0 MB/s eta 0:00:00
_____ 52.0/52.0 kB 5.3 MB/s eta 0:00:00
_____ 44.0/44.0 kB 4.7 MB/s eta 0:00:00
_____ 564.3/564.3 kB 31.1 MB/s eta 0:00:00
_____ 24.1/24.1 MB 27.7 MB/s eta 0:00:00
_____ 1.0/1.0 MB 96.4 MB/s eta 0:00:00
_____ 2.5/2.5 MB 148.6 MB/s eta 0:00:00
_____ 19.9/19.9 MB 168.7 MB/s eta 0:00:00
_____ 486.6/486.6 kB 52.8 MB/s eta 0:00:00
_____ 19.2/19.2 MB 136.1 MB/s eta 0:00:00
_____ 278.2/278.2 kB 39.1 MB/s eta 0:00:00
_____ 2.0/2.0 MB 158.7 MB/s eta 0:00:00
_____ 458.9/458.9 kB 59.1 MB/s eta 0:00:00
_____ 17.2/17.2 MB 275.3 MB/s eta 0:00:00
_____ 72.1/72.1 kB 10.9 MB/s eta 0:00:00
_____ 142.0/142.0 kB 22.1 MB/s eta 0:00:00
_____ 68.7/68.7 kB 10.1 MB/s eta 0:00:00
_____ 231.6/231.6 kB 31.9 MB/s eta 0:00:00
_____ 66.5/66.5 kB 8.7 MB/s eta 0:00:00
_____ 105.4/105.4 kB 15.0 MB/s eta 0:00:00
_____ 71.6/71.6 kB 10.3 MB/s eta 0:00:00
_____ 60.6/60.6 kB 7.5 MB/s eta 0:00:00
_____ 64.7/64.7 kB 7.6 MB/s eta 0:00:00
_____ 12.0/12.0 MB 248.8 MB/s eta 0:00:00
_____ 51.0/51.0 kB 6.8 MB/s eta 0:00:00

```

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

google-colab 1.0.0 requires requests==2.32.4, but you have requests 2.32.5 which is incompatible.

opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-exporter-otlp-proto-common==1.38.0, but you have opentelemetry-exporter-otlp-proto-common 1.40.0 which is incompatible.

opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-proto==1.38.0, but you have opentelemetry-proto 1.40.0 which is incompatible.

opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-sdk~=1.38.0, but you have opentelemetry-sdk 1.40.0 which is incompatible.

opentelemetry-exporter-gcp-logging 1.11.0a0 requires opentelemetry-sdk<1.39.0,>=1.35.0, but you have opentelemetry-sdk 1.40.0 which is incompatible.

google-adk 1.27.1 requires opentelemetry-api<1.39.0,>=1.36.0, but you have opentelemetry-api 1.40.0 which is incompatible.

google-adk 1.27.1 requires opentelemetry-sdk<1.39.0,>=1.36.0, but you have opentelemetry-sdk 1.40.0 which is incompatible.

langgraph-prebuilt 1.0.8 requires langchain-core>=1.0.0, but you have langchain-core 0.3.83 which is incompatible.

Note:

- After running the above cell, kindly restart the runtime (for Google Colab) or notebook kernel (for Jupyter Notebook), and run all cells sequentially from the next cell.
- On executing the above line of code, you might see a warning regarding package dependencies. This error message can be ignored as the above code ensures that all necessary libraries and their dependencies are maintained to successfully execute the code in **this notebook**.

```

#Libraries for processing dataframes,text
import json,os
import tiktoken

```

```
import pandas as pd

#Libraries for Loading Data, Chunking, Embedding, and Vector Databases
from langchain.text_splitter import RecursiveCharacterTextSplitter
from langchain_community.document_loaders import PyMuPDFLoader
from langchain_community.embeddings.sentence_transformer import SentenceTransformer
from langchain_community.vectorstores import Chroma

#Libraries for downloading and loading the llm
from huggingface_hub import hf_hub_download
from llama_cpp import Llama
```

Question Answering using LLM

Downloading and Loading the model

```
model_name_or_path = "TheBloke/Mistral-7B-Instruct-v0.2-GGUF"
model_basename = "mistral-7b-instruct-v0.2.Q6_K.gguf"
```

```
model_path = hf_hub_download(
    repo_id=model_name_or_path,
    filename=model_basename
)
```

```
mistral-7b-instruct-v0.2.Q6_K.gguf: 0% | 0.00/5.94G [00:00<?, ?B/s]
```

```
#uncomment the below snippet of code if the runtime is connected to GPU.
llm = Llama(
    model_path=model_path,
    n_ctx=7000,
    n_gpu_layers=36,
    n_batch=512
)
```

```
AVX = 1 | AVX_VNNI = 1 | AVX2 = 1 | AVX512 = 1 | AVX512_VBMI = 1 | AVX512_VNNI = 1
| FMA = 1 | NEON = 0 | ARM_FMA = 0 | F16C = 1 | FP16_VA = 0 | WASM_SIMD = 0 | BLAS
= 1 | SSE3 = 1 | SSSE3 = 1 | VSX = 0 |
```

```
#Observation : The model path is configured to TheBloke/Mistral-7B-Instruct-v0.2-GGUF
#A 5.94GB files is downloaded from hugging face.
#The Llama class is instantiated to load the model into memory.
```

Response

```
def response(query,max_tokens=128,temperature=0,top_p=0.95,top_k=50):
    model_output = llm(
        prompt=query,
        max_tokens=max_tokens,
        temperature=temperature,
        top_p=top_p,
        top_k=top_k
    )

    return model_output['choices'][0]['text']
```

```
response("What treatment options are available for managing hypertension?")
```

'\n\nHypertension, or high blood pressure, is a common condition that can increase the risk of various health problems such as heart disease, stroke, and kidney damage. The good news is that there are several effective treatment options available to help manage hypertension and reduce the risk of complications. Here are some of the most commonly used treatments:\n\n1. Lifestyle modifications: Making lifestyle changes is often the first line of defense against hypertension. This may include eating a healthy diet rich in fruits, vegetables, whole grains, and lean proteins; limiting sodium intake; getting regular physical activity'

#Observation : A general answer for the query is given by the LLM.

Query 1: What is the protocol for managing sepsis in a critical care unit?

```
user_input_1 = "What is the protocol for managing sepsis in a critical care unit?"  
response(user_input_1)
```

Llama.generate: prefix-match hit

'\n\nSepsis is a life-threatening condition that can arise from an infection, and it requires prompt recognition and aggressive management in a critical care unit. The following are general steps for managing sepsis in a critical care unit:\n\n1. Early recognition: Recognize the signs and symptoms of sepsis early and initiate treatment as soon as possible. Sepsis can present with various clinical features, including fever or hypothermia, tachycardia or bradycardia, altered mental status, respiratory distress, and lactic acidosis.\n2. ABCs'

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
user_input_2 = "What are the common symptoms for appendicitis, and can it be cured?"  
response(user_input_2)
```

Llama.generate: prefix-match hit

'\n\nAppendicitis is a medical condition characterized by inflammation of the appendix, a small tube-shaped organ located in the lower right side of the abdomen. The symptoms of appendicitis can vary from person to person, but some common signs and symptoms include:\n\n1. Abdominal pain: The pain is typically located in the lower right quadrant of the abdomen and may be sharp, crampy, or dull. It may start as a mild discomfort that comes and goes, but it usually worsens over time and becomes constant and severe.\n2. Loss of appetite'

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
user_input_3 = "What are the effective treatments or solutions for addressing sudden patchy hair loss?"  
response(user_input_3)
```

Llama.generate: prefix-match hit

'\n\nSudden patchy hair loss, also known as alopecia areata, is an autoimmune condition that affects the hair follicles. In this condition, the immune system mistakenly attacks the hair follicles, causing hair to stop growing or fall out in small patches. The exact cause of alopecia areata is not known, but it is believed to be related to genetics and environmental factors.\n\nThere are several treatments that can help address sudden patchy hair loss:\n\n1. Corticosteroids: These medications can be applied directly to the affected area or taken orally'

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
user_input_4 = "What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?"
response(user_input_4)
```

Llama.generate: prefix-match hit

'\n\nA person who has sustained a physical injury to the brain tissue may require various treatments depending on the severity and location of the injury. Here are some common treatments that may be recommended:\n\n1. Emergency care: In case of a traumatic brain injury (TBI), it is essential to seek emergency medical attention as soon as possible. The primary goal of emergency care is to prevent further damage to the brain, stabilize vital signs, and manage any life-threatening conditions.\n\n2. Medications: Depending on the symptoms, healthcare professionals may prescribe medications to manage various conditions associated with a brain injury, such as pain, swelling, and seizures.'

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
user_input_5 = "What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?"
response(user_input_5)
```

Llama.generate: prefix-match hit

"\n\nFirst and foremost, if you suspect that someone has fractured their leg while hiking, it's essential to ensure their safety and prevent further injury. Here are some necessary precautions:\n\n1. Keep the person calm and still: Encourage them to remain as still as possible to minimize pain and prevent worsening the injury.\n\n2. Assess the situation: Check for any signs of shock, such as pale skin, rapid heartbeat, or shallow breathing. If you notice these symptoms, seek medical help immediately.\n\n3. Immobilize the leg: Use a splint, sling, or a makeshift splint made of wood or cardboard to stabilize the leg and prevent movement.'

#Observation : For the query 1 to 5, The model provides generic medical response.

Question Answering using LLM with Prompt Engineering

```
system_prompt = "You are a helpful medical assistant. Provide accurate and concise answers to medical queries."
response(system_prompt)
```

Query 1: What is the protocol for managing sepsis in a critical care unit?

```
user_input_1 = system_prompt + "\n" + "What is the protocol for managing sepsis in a critical care unit?"
response(user_input_1)
```

'\nSepsis is a life-threatening condition caused by a dysregulated response to infection. In a critical care unit, managing sepsis involves addressing the underlying infection while providing supportive care to maintain organ function and prevent progression to septic shock. Here are some key steps in managing sepsis in a critical care unit based on current medical guidelines:\n1. Early recognition and diagnosis: Recognize sepsis early by assessing for signs of infection, inflammation, and organ dysfunction. Use the Sequential Organ Failure Assessment (SOFA) score to evaluate organ function and the Quick'

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
user_input_2 = system_prompt+"\n"+ " What are the common symptoms for appendicitis,
response(user_input_2)
```

Llama.generate: prefix-match hit

'\n\nAppendicitis is an inflammatory condition of the appendix, a small tube-shaped organ located in the lower right side of the abdomen. Common symptoms include:\n1. Sudden and persistent pain in the lower right abdomen, often starting around the navel and moving to the right.\n2. Loss of appetite and feeling sick to your stomach (nausea).\n3. Fever, which may be low-grade at first but can rise as high as 101°F (38.3°C) or more.\n4. Vomiting,'

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
user_input_3 = system_prompt+"\n"+ " What are the effective treatments or solutions
response(user_input_3)
```

Llama.generate: prefix-match hit

"\n\nSudden patchy hair loss, also known as alopecia areata, is an autoimmune condition that results in the sudden appearance of smooth, round patches of baldness on the scalp or other areas of the body. The exact cause of alopecia areata is not known, but it's believed to be related to a problem with the immune system.\n\nThere are several treatments for addressing sudden patchy hair loss:\n\n1. Corticosteroids: These medications can be applied directly to the affected area or taken orally to reduce inflammation and suppress the immune response"

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
user_input_4 = system_prompt+"\n"+ "What treatments are recommended for a person wh
response(user_input_4)
```

Llama.generate: prefix-match hit

'\n\nAccording to the American Association of Neurological Surgeons (AANS) and the Congress of Neurological Surgeons (CNS), the treatment for a brain injury depends on the severity and location of the injury. Here are some common treatments:\n1. Mild Brain Injuries: For mild brain injuries, also known as concussions, rest is the best treatment. The person should avoid physical activity, alcohol, and caffeine. They may need to take over-the-counter pain relievers for headaches and should follow up with their healthcare provider if symptoms persist'

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
user_input_5 = system_prompt+"\n"+ "What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?"
response(user_input_5)
```

Llama.generate: prefix-match hit

'\nA fractured leg, also known as a broken leg, is a common injury that can occur during hiking due to falls or accidents. Here are the necessary precautions and treatment steps for a person who has sustained a fractured leg:\n1. Assess the situation: Ensure the safety of both the injured person and those around them. Check for any signs of shock, such as pale skin, rapid heartbeat, or shallow breathing. If there is any suspicion of a severe injury or if the person shows signs of shock, call emergency services immediately.\n2. Immobilize the leg: Use a splint or'

#Observation : All the answers are based on the internal training of the model, generated by the model. As we have defined in the system prompt that it should respond like a medical assistant, the model is referring American Speech-Language and Hearing (ASHA) guidelines.

Question Answering using LLM with Prompt Engineering - With the Different System Prompt

```
system_prompt_1 = "Act as a nurse and outline possible next steps."
```

Query 1: What is the protocol for managing sepsis in a critical care unit?

```
user_input_1 = system_prompt_1+"\n"+ "What is the protocol for managing sepsis in a critical care unit?"
response(user_input_1)
```

Llama.generate: prefix-match hit

" Sepsis is a life-threatening condition that requires immediate intervention. Here are the possible next steps for managing sepsis in a critical care unit:\n1. Assess the patient's vital signs, level of consciousness, and overall clinical status.\n2. Administer oxygen therapy to maintain adequate oxygenation.\n3. Initiate intravenous fluid resuscitation with isotonic crystalloids such as normal saline or lactated Ringer's solution. The goal is to achieve a mean arterial pressure (MAP) of 65 mmHg or higher and a central venous pressure (CVP) of 8-12 mmHg."

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
user_input_2 = system_prompt_1+"\n"+ "What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?"
response(user_input_2)
```

Llama.generate: prefix-match hit


```
'\n\nSymptoms of appendicitis may include:\n1. Sudden pain in the lower right abdomen that begins near the navel and then shifts to the right side.\n2. Loss of appetite.\n3. Nausea and vomiting.\n4. Fever, which may start after a few hours of having abdominal pain.\n5. Abdominal swelling and rigidity.\n6. Inability to pass gas or have a bowel movement.\n7. Diarrhea or constipation.\n8. Pain upon walking, coughing, or making other'
```

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
user_input_3 = system_prompt_1+"\n"+ " What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?"
response(user_input_3)
```

Llama.generate: prefix-match hit

```
"\nFirstly, it's important to note that sudden patchy hair loss can have various causes, some of which may require medical attention. Here are some possible next steps:\n1. Consult a healthcare professional or dermatologist: A healthcare professional or dermatologist can help determine the underlying cause of the hair loss and recommend appropriate treatments. They may perform blood tests, scalp biopsies, or other diagnostic tests to rule out any medical conditions that could be contributing to the hair loss.\n2. Rule out stress-related hair loss: Stress can cause sudden patchy hair loss, known as telogen"
```

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
user_input_4 = system_prompt_1+"\n"+ "What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?"
response(user_input_4)
```

Llama.generate: prefix-match hit

```
" The treatment plan for a person with a brain injury can vary depending on the severity and location of the injury. Here are some common interventions that may be recommended:\n1. Medical stabilization: The first priority is to ensure the person's vital signs are stable and they receive appropriate medical care for any life-threatening conditions, such as bleeding or swelling in the brain.\n2. Rehabilitation: Depending on the extent of the injury, rehabilitation may be necessary to help the person regain lost function and improve their overall quality of life. This can include physical therapy to help with mobility,"
```

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
user_input_5 = system_prompt_1+"\n"+ "What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?"
response(user_input_5)
```

Llama.generate: prefix-match hit

```
'\n\nFirst and foremost, it is essential to ensure the safety and stability of the injured person. If the fracture is obvious or suspected, avoid moving them unnecessarily to prevent further injury or discomfort. Here are some possible next steps:\n\n1. Assess the severity of the injury: Determine if the fracture is open (compound) or closed (simple). Open fractures require immediate medical attention due to the risk of infection.\n2. Apply a splint or immobilize the leg: Use a makeshift splint, such as a stick or hiking pole, to stabilize'
```

```
#observation : All the answers are based on the internal training of the model, generated by the system prompt. As the system prompt is changed and mentioned to act like a nurse and outline the steps, all the response contains a small description about the medical condition and the steps.
```

Data Preparation for RAG

```
#Libraries for processing dataframes, text
import json, os
import tiktoken
import pandas as pd

#Libraries for Loading Data, Chunking, Embedding, and Vector Databases
from langchain.text_splitter import RecursiveCharacterTextSplitter
from langchain_community.document_loaders import PyMuPDFLoader
from langchain_community.embeddings.sentence_transformer import SentenceTransformer
from langchain_chroma import Chroma
```

Loading the Data

```
# uncomment and run the below code snippets if the dataset is present in the Google Drive
# from google.colab import drive
# drive.mount('/content/drive')
```

```
manual_pdf_path = "/content/medical_diagnosis_manual.pdf"
```

```
pdf_loader = PyMuPDFLoader>manual_pdf_path)
```

```
manual = pdf_loader.load()
```

```
#observation: The Merck Manual pdf is loaded successfully
```

Data Overview

Checking the first 5 pages

```
for i in range(5):
    print(f"Page Number : {i+1}", end="\n")
    print>manual[i].page_content, end="\n")
```

Page Number : 1
sangeethakumari.g@gmail.com
D83YNKBVRA

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470	Chapter 42. Neck & Back Pain
481	Chapter 43. Hand Disorders
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Chapter 93. Thyroid Disorders

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.....
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.....
.....
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```
#Observation : The pdf is loaded and we can see the first few pages.
```

Checking the number of pages

```
len(manual)
```

```
4114
```

```
#Observation : this is the total number of pages in the medical manual loaded.
```

Data Chunking

```
text_splitter = RecursiveCharacterTextSplitter.from_tiktoken_encoder(  
    encoding_name='cl100k_base',  
    chunk_size=512,  
    chunk_overlap=20  
)
```

```
#OBSERVATION : Defining the chunk size as 512 and the chunk overlap size as 20.  
#The text splitter has been successfully initialized with a chunk size of 512 and a
```

```
document_chunks = pdf_loader.load_and_split(text_splitter)
```

```
len(document_chunks)
```

```
8492
```

```
#OBSERVATION : The number of chunks are 8492
```

```
document_chunks[0].page_content
```

```
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ll is liable for legal action.'
```

```
document_chunks[1].page_content
```

```
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angeethakumari.g@gmail.com only.\nSharing or publishing the contents in part or fu  
ll is liable for legal action.'
```

```
document_chunks[2].page_content
```

'Table of Contents\n1\nFront

.....	
.....	
.....\n1\nCover	
.....	
.....\n2\nFront Matter	
.....	
.....\n53\n1 – Nutritional Disorders	
.....\n53	
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.....\n59\nChapter 2. Undernutrition	
.....\n69\nChapter 3. Nutritional Support	
.....\n76\nChapter 4. Vitamin Deficiency, Dependency & Toxicity	
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.....\n120\n2 – Gastrointestinal Disorders	
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.....\n225\nChapter 18. Irritable Bowel Syndrome	
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.....\n246\nChapter 21. Anorectal Disorders
.....\n254\nChapter 22. Tumors of the GI Tract
.....\n275\n3 - Hepatic & Biliary Disorders
.....\n275\nChapter 23. Approach to the Patient With Liver Disease
.....\n294\nChapter 24. Testing for Hepatic & Biliary Disorders
.....\n305'

document_chunks[3].page_content

"Chapter 24. Testing for Hepatic & Biliary Disorders
.....
.....\n305\nChapter 25. Drugs & the Liver
.....
.....\n308\nChapter 26. Alcoholic Liver Disease
.....
.....\n314\nChapter 27. Fibrosis & Cirrhosis
.....
.....\n322\nChapter 28. Hepatitis
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.....
.....\n463\nChapter 41. Bursa, Muscle & Tendon Disorders
.....
.....\n470\nChapter 42. Neck & Back Pain
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.....\n481\nChapter 43. Hand Disorders
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Last 200 characters of `document_chunks[2]` :

```
print(document_chunks[2].page_content[-200:])
```

```
.....
294
Chapter 24. Testing for Hepatic & Biliary Disorders
.....
.....
305
```

First 200 characters of `document_chunks[3]` :

```
print(document_chunks[3].page_content[:200])
```

```
Chapter 24. Testing for Hepatic & Biliary Disorders
.....
.....
305
Chapter 25. Drugs & the Liver      .....
```

#OBSERVATION : As expected, there are some overlaps:

*#The sentence 'Chapter 24. Testing for Hepatic & Biliary Disorders
#If we increase the chunk_overlap, the overlapping length of the sentence will also*

Data Chunking - Changing the chunk size and chunk_overlap parameter

```
text_splitter = RecursiveCharacterTextSplitter.from_tiktoken_encoder(
    encoding_name='cl100k_base',
    chunk_size=1024,
    chunk_overlap=20
)
```

*#OBSERVATION : Defining the chunk size as 1024 and the chunk overlap size as 20.
#The text splitter has been successfully initialized with a chunk size of 1024 and*

```
document_chunks = pdf_loader.load_and_split(text_splitter)
```

```
len(document_chunks)
```

```
4558
```

*# Observation: Increasing the chunk size from 512 to 1024 tokens resulted in a sign
#This reduces the overhead for the vector database during the similarity search pro*

```
document_chunks[100].page_content
```

"particularly in patients with alcoholism or protein-energy undernutrition. Diagnosis is usually clinical. There is no single accepted laboratory test of vitamin B6 status; measurement of serum pyridoxal phosphate is most common. Treatment

- Pyridoxine
- Elimination of risk factors when possible

For secondary deficiency, causes (eg, use of pyridoxine-inactivating drugs, malabsorption) should be corrected if possible. Usually, pyridoxine 50 to 100 mg po once/day corrects the deficiency in adults. Most people taking isoniazid should also be given pyridoxine 30 to 50 mg/day. For deficiency due to increased metabolic demand, amounts larger than the daily recommended intake may be required. For most cases of inborn errors of metabolism, high doses of pyridoxine may be effective.

Vitamin B6 Toxicity

The ingestion of megadoses (> 500 mg/day) of pyridoxine (eg, taken to treat carpal tunnel syndrome or premenstrual syndrome although efficacy is unproved) may cause peripheral neuropathy with deficits in a stocking-glove distribution, including progressive sensory ataxia and severe impairment of position and vibration senses. Senses of touch, temperature, and pain are less affected. Motor and central nervous systems are usually intact.

Diagnosis is clinical. **Treatment** is to stop taking vitamin B6. Recovery is slow and, for some patients, incomplete.

Vitamin B12

Cobalamin is a general term for compounds with biologic vitamin B12 activity. These compounds are involved in nucleic acid metabolism, methyl transfer, and myelin synthesis and repair. They are necessary for the formation of normal RBCs.

Food-bound vitamin B12 is released in the stomach's acid environment and is bound to R protein (haptocorrin). Pancreatic enzymes cleave this B12 complex (B12-R protein) in the small intestine. After cleavage, intrinsic factor, secreted by parietal cells in the gastric mucosa, binds with vitamin B12. Intrinsic factor is required for absorption of vitamin B12, which takes place in the terminal ileum.

Vitamin B12 in plasma is bound to transcobalamins I and II. Transcobalamin II is responsible for delivering vitamin B12 to tissues. The liver stores large amounts of vitamin B12. Enterohepatic reabsorption helps retain vitamin B12. Liver vitamin B12 stores can normally sustain physiologic needs for 3 to 5 yr if B12 intake stops (eg, in people who become vegans) and for months to 1 yr if enterohepatic reabsorption capacity is absent.

Large amounts of vitamin B12 seem to be nontoxic but are not recommended for regular use (ie, as a general tonic).

Vitamin B12 Deficiency

Dietary vitamin B12 deficiency usually results from inadequate absorption, but deficiency can develop in vegans who do not take vitamin supplements. Deficiency causes megaloblastic anemia, damage to the white matter of the spinal cord and brain, and peripheral neuropathy.

Diagnosis is usually made by measuring serum vitamin B12 levels. The Schilling test helps determine etiology. **Treatment** consists of oral or parenteral vitamin B12. Folate (folic acid) should not be used instead of vitamin B12 because folate may alleviate the anemia but allow

The Merck Manual of Diagnosis & Therapy, 19th Edition Chapter 4. Vitamin Deficiency, Dependency & Toxicity

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"neurologic deficits to progress.\nEtiology\nInadequate vitamin B12 intake is possible in vegans but is otherwise unlikely. Breastfed babies of vegan\nmothers may develop vitamin B12 deficiency by age 4 to 6 mo because their\n[\nTable 4–5. Causes of Vitamin B12 Deficiency]\nliver stores (which are normally extensive) are limited and their rapid growth rate results in high demand.\nVitamin B12 deficiency usually results from inadequate absorption (see Table 4–5 and p. 153), which, in\nthe elderly, most commonly results from decreased acid secretion. In such cases, crystalline vitamin B12\n(such as that available in vitamin supplements) can be absorbed, but food-bound vitamin B12 is not\nliberated and absorbed normally. Inadequate absorption may occur in blind loop syndrome (with\novergrowth of bacteria) or fish tapeworm infestation; in these cases, bacteria or parasites use ingested\nvitamin B12 so that less is available for absorption. Vitamin B12 absorption may be inadequate if ileal\nabsorptive sites are destroyed by inflammatory bowel disease or are surgically removed. Less common\ncauses of inadequate vitamin B12 absorption include chronic pancreatitis, gastric surgery, malabsorption\nsyndromes, AIDS, use of certain drugs (eg, antacids, metformin), repeated exposure to nitrous oxide, and\na genetic disorder causing malabsorption in the ileum (Imerslund–Graesbeck syndrome).\nPernicious anemia is often used synonymously with vitamin B12 deficiency. However, pernicious\nanemia specifically refers to anemia resulting from vitamin B12 deficiency caused by an autoimmune\nmetaplastic atrophic gastritis with loss of intrinsic factor (see p. 133). Patients with classic pernicious\nanemia, most commonly younger adults, are at increased risk of stomach and other GI cancers.\nSubacute combined degeneration refers to degenerative changes in the nervous system due to\nvitamin B12 deficiency; they affect mostly brain and spinal cord white matter. Demyelinating or axonal\nperipheral neuropathies can occur.\nSymptoms and Signs\nAnemia usually develops insidiously. It is often more severe than its symptoms indicate because its slow\nevolution allows physiologic adaptation. Occasionally, splenomegaly and hepatomegaly occur. Various GI\nsymptoms, including weight loss and poorly localized abdominal pain, may occur. Glossitis, usually\ndescribed as burning of the tongue, is uncommon.\nNeurologic symptoms develop independently from and often without hematologic abnormalities. In early\nstages, decreased position and vibratory sensation in the extremities is accompanied by mild to moderate\nweakness and hyporeflexia. In later stages, spasticity, extensor plantar responses, greater loss of position\nand vibratory sensation in the lower extremities, and ataxia emerge. These deficits may develop in a\nstocking-glove distribution. Tactile, pain, and temperature sensations are usually spared but may be\ndifficult to assess in the elderly.\nSome patients are also irritable and mildly depressed. Paranoia (megaloblastic madness), delirium,\nconfusion, spastic ataxia, and, at times, postural hypotension may occur in advanced cases. The\nconfusion may be difficult to differentiate from age-related dementias, such as Alzheimer's disease.\nDiagnosis\n• CBC and vitamin B12 and folate levels\n• Sometimes methylmalonic acid levels or Schilling test\nThe Merck Manual of Diagnosis & Therapy, 19th Edition Chapter 4. Vitamin Deficiency, Dependency & Toxicity\n87\nsangeethakumari.g@gmail.com\nD83YNKBVRA\nThis file is meant for personal use by sangeethakumari.g@gmail.com only.\nSharing or publishing the contents in part or full is liable for legal action."

*#Observation : Larger chunks provide the LLM with more surrounding context in a single chunk.
For complex medical topics like Vitamin B12 deficiency, this helps the model understand the symptoms and treatments without needing to 'stitch together' information from multiple separate chunks.*

```
from langchain_community.embeddings.sentence_transformer import SentenceTransformer
```

Embedding

```
embedding_model = SentenceTransformerEmbeddings(model_name="all-MiniLM-L6-v2")
```

```
/tmp/ipykernel_8942/3002436811.py:1: LangChainDeprecationWarning: The class `HuggingFaceEmbeddings` was deprecated in LangChain 0.2.2 and will be removed in 1.0. An updated version of the class exists in the :class:`~langchain-huggingface` package and should be used instead. To use it run `pip install -U :class:`~langchain-huggingface` and import as `from :class:`~langchain_huggingface import HuggingFaceEmbeddings`.
```

```
embedding_model = SentenceTransformerEmbeddings(model_name="all-MiniLM-L6-v2")
modules.json: 0%|          | 0.00/349 [00:00<?, ?B/s]
config_sentence_transformers.json: 0%|          | 0.00/116 [00:00<?, ?B/s]
README.md: 0.00B [00:00, ?B/s]
sentence_bert_config.json: 0%|          | 0.00/53.0 [00:00<?, ?B/s]
config.json: 0%|          | 0.00/612 [00:00<?, ?B/s]
model.safetensors: 0%|          | 0.00/90.9M [00:00<?, ?B/s]
tokenizer_config.json: 0%|          | 0.00/350 [00:00<?, ?B/s]
vocab.txt: 0.00B [00:00, ?B/s]
tokenizer.json: 0.00B [00:00, ?B/s]
special_tokens_map.json: 0%|          | 0.00/112 [00:00<?, ?B/s]
config.json: 0%|          | 0.00/190 [00:00<?, ?B/s]
```

```
#Observation : The SentenceTransformerEmbeddings model (specifically the all-MiniLM
```

```
embedding_1 = embedding_model.embed_query(document_chunks[0].page_content)
embedding_2 = embedding_model.embed_query(document_chunks[1].page_content)
```

```
print("Dimension of the embedding vector ",len(embedding_1))
len(embedding_1)==len(embedding_2)
```

```
Dimension of the embedding vector 384
```

```
True
```

```
#OBSERVATION :
```

```
# The embedding model provides a fixed-length vector for any number of chunks.
# This is necessary because we want to compare them for similarity.
```

```
embedding_1,embedding_2
```

([-0.1213507428765297,
0.061943452805280685,
0.013677269220352173,
-0.07614820450544357,
0.06022114306688309,
-0.022122623398900032,
0.027555733919143677,
0.014887353405356407,
0.028904058039188385,
0.061900440603494644,
0.03526139631867409,
0.048970114439725876,
0.05497710034251213,
-0.07574319839477539,
-0.024668468162417412,
0.0026807691901922226,
-0.06615706533193588,
-0.02556614577770233,
-0.07604912668466568,
0.013985793106257915,
-0.029712015762925148,
0.009356653317809105,
0.002295687096193433,
0.001890511717647314,
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-0.04808863624930382,
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-0.03497074171900749,
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-0.006988671142607927,
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-0.016004793345928192,
-0.04776211455464363,
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-0.07931889593601227,
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-0.027515124529600143,
-0.12843795120716095,
0.03427319601178169,
0.04356931149959564,
0.005167060531675816,

0.003391595557332039,
-0.054239265620708466,
-0.006427630316466093,
-0.012221387587487698,
-0.07941500842571259,
-0.14338813722133636,
0.09406856447458267,
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-0.024407681077718735,  
0.05750381574034691,  
0.0011883415281772614,  
0.1002439484000206,  
-0.04319382458925247,  
0.018191756680607796,  
-0.018964454531669617,  
-0.03668832406401634,  
0.0064648790284991264])
```

Vector Database

```
out_dir = 'medical_db'  
  
if not os.path.exists(out_dir):  
    os.makedirs(out_dir)
```

```
vectorstore = Chroma.from_documents(  
    document_chunks,  
    embedding_model,
```

```
)
    persist_directory=out_dir
```

```
vectorstore = Chroma(persist_directory=out_dir,embedding_function=embedding_model)
```

```
# Observation : All your processed text chunks are converted into a numerical embedding  
# and then stored as embeddings (along with their original text content) into a new
```

```
vectorstore.embeddings
```

```
HuggingFaceEmbeddings(client=SentenceTransformer(  
  (0): Transformer({'max_seq_length': 256, 'do_lower_case': False, 'architecture':  
    'BertModel'})  
  (1): Pooling({'word_embedding_dimension': 384, 'pooling_mode_cls_token': False,  
    'pooling_mode_mean_tokens': True, 'pooling_mode_max_tokens': False, 'pooling_mode_  
    mean_sqrt_len_tokens': False, 'pooling_mode_weightedmean_tokens': False, 'pooling_  
    mode_lasttoken': False, 'include_prompt': True})  
  (2): Normalize()  
, model_name='all-MiniLM-L6-v2', cache_folder=None, model_kwargs={}, encode_kwar  
gs={}, multi_process=False, show_progress=False)
```

```
#Observation: This code shows the configuration of the SentenceTransformerEmbedding  
# that is responsible for converting your text data into numerical vector represent
```

```
vectorstore.similarity_search("What are the common symptoms for appendicitis, and c
```

[Document(id='7b0f69cc-5274-4e5d-b036-72959bfd5de2', metadata={'keywords': '', 'trapped': '', 'moddate': '2026-03-18T21:26:23+00:00', 'title': 'The Merck Manual of Diagnosis & Therapy, 19th Edition', 'author': '', 'format': 'PDF 1.7', 'creationdate': '2012-06-15T05:44:40+00:00', 'file_path': '/content/medical_diagnosis_manual.pdf', 'subject': '', 'modDate': 'D:20260318212623Z', 'source': '/content/medical_diagnosis_manual.pdf', 'page': 172, 'producer': 'pdf-lib (https://github.com/Hopding/pdf-lib)', 'creationDate': 'D:20120615054440Z', 'creator': 'Atop CHM to PDF Converter', 'total_pages': 4114}, page_content="Appendicitis\nAppendicitis is acute inflammation of the vermiform appendix, typically resulting in abdominal pain, anorexia, and abdominal tenderness. Diagnosis is clinical, often supplemented by CT or ultrasound. Treatment is surgical removal.\nIn the US, acute appendicitis is the most common cause of acute abdominal pain requiring surgery. Over 5% of the population develops appendicitis at some point. It most commonly occurs in the teens and 20s but may occur at any age.\nOther conditions affecting the appendix include carcinoids, cancer, villous adenomas, and diverticula. The appendix may also be affected by Crohn's disease or ulcerative colitis with pancolitis.\nThe Merck Manual of Diagnosis & Therapy, 19th Edition\nChapter 11. Acute Abdomen & Surgical Gastroenterology\n163\nsangeethakumari.g@gmail.com\nD83YNKBVRA\nThis file is meant for personal use by sangeethakumari.g@gmail.com only.\nSharing or publishing the contents in part or full is liable for legal action."),

Document(id='75eb5fd3-83ca-44c4-91c9-1840d02da6fc', metadata={'creator': 'Atop CHM to PDF Converter', 'format': 'PDF 1.7', 'creationdate': '2012-06-15T05:44:40+00:00', 'source': '/content/medical_diagnosis_manual.pdf', 'subject': '', 'modDate': 'D:20260318212623Z', 'total_pages': 4114, 'author': '', 'file_path': '/content/medical_diagnosis_manual.pdf', 'moddate': '2026-03-18T21:26:23+00:00', 'keywords': '', 'creationDate': 'D:20120615054440Z', 'trapped': '', 'title': 'The Merck Manual of Diagnosis & Therapy, 19th Edition', 'page': 173, 'producer': 'pdf-lib (https://github.com/Hopding/pdf-lib)'}, page_content="Etiology\nAppendicitis is thought to result from obstruction of the appendiceal lumen, typically by lymphoid hyperplasia, but occasionally by a fecalith, foreign body, or even worms. The obstruction leads to distention, bacterial overgrowth, ischemia, and inflammation. If untreated, necrosis, gangrene, and perforation occur. If the perforation is contained by the omentum, an appendiceal abscess results.\nSymptoms and Signs\nThe classic symptoms of acute appendicitis are epigastric or periumbilical pain followed by brief nausea, vomiting, and anorexia; after a few hours, the pain shifts to the right lower quadrant. Pain increases with cough and motion. Classic signs are right lower quadrant direct and rebound tenderness located at McBurney's point (junction of the middle and outer thirds of the line joining the umbilicus to the anterior superior spine). Additional signs are pain felt in the right lower quadrant with palpation of the left lower quadrant (Rovsing sign), an increase in pain from passive extension of the right hip joint that stretches the iliopsoas muscle (psoas sign), or pain caused by passive internal rotation of the flexed thigh (obturator sign). Low-grade fever (rectal temperature 37.7 to 38.3° C [100 to 101° F]) is common.\nUnfortunately, these classic findings appear in < 50% of patients. Many variations of symptoms and signs occur. Pain may not be localized, particularly in infants and children. Tenderness may be diffuse or, in rare instances, absent. Bowel movements are usually less frequent or absent; if diarrhea is a sign, a retrocecal appendix should be suspected. RBCs or WBCs may be present in the urine. Atypical symptoms are common among elderly patients and pregnant women; in particular, pain is less severe and local tenderness is less marked.\nDiagnosis\n• Clinical evaluation\n• Abdominal CT if necessary\n• Ultrasound an option to CT\nWhen classic symptoms and signs are present, the diagnosis is clinical. In such patients, delaying laparotomy to do imaging tests only increases the likelihood of perforation and subsequent complications.\nIn patients with atypical or equivocal findings, imaging studies should be done without delay. Contrast-"),

Document(id='1eddd0491-de65-41df-a6c8-e89f0ca901de', metadata={'creationDate': 'D:20120615054440Z', 'total_pages': 4114, 'page': 173, 'file_path': '/content/medical_diagnosis_manual.pdf', 'creator': 'Atop CHM to PDF Converter', 'source': '/content/medical_diagnosis_manual.pdf', 'modDate': 'D:20260318212623Z', 'trapped': '', 'author': '', 'keywords': '', 'subject': '', 'creationdate': '2012-06-15T05:44:40+00:00', 'creationDate': 'D:20120615054440Z', 'creator': 'Atop CHM to PDF Converter', 'source': '/content/medical_diagnosis_manual.pdf', 'modDate': 'D:20260318212623Z', 'trapped': '', 'author': '', 'keywords': '', 'subject': ''}),

```
0:00', 'format': 'PDF 1.7', 'producer': 'pdf-lib (https://github.com/Hopding/pdf-lib)', 'title': 'The Merck Manual of Diagnosis & Therapy, 19th Edition', 'moddate': '2026-03-18T21:26:23+00:00'}, page_content='enhanced CT has reasonable accuracy in diagnosing appendicitis and can also reveal other causes of\nan acute abdomen. Graded compression ultrasound can usually be done quickly and uses no radiation\n(of particular concern in children); however, it is occasionally limited by the presence of bowel gas and is\nless useful for recognizing nonappendiceal causes of pain. Appendicitis remains primarily a clinical\ndiagnosis. Selective and judicious use of radiographic studies may reduce the rate of negative\nlaparotomy.\nLaparoscopy can be used for diagnosis as well as definitive treatment; it may be especially helpful in\nwomen with lower abdominal pain of unclear etiology. Laboratory studies typically show leukocytosis\n(12,000 to 15,000/ $\mu$ L), but this finding is highly variable; a normal WBC count should not be used to\nexclude appendicitis.\nPrognosis\nWithout surgery or antibiotics, mortality is > 50%.\nWith early surgery, the mortality rate is < 1%, and convalescence is normally rapid and complete. With\ncomplications (rupture and development of an abscess or peritonitis), the prognosis is worse: Repeat\noperations and a long convalescence may follow.\nTreatment\nThe Merck Manual of Diagnosis & Therapy, 19th Edition\nChapter 11. Acute Abdomen & Surgical Gastroenterology\n164\nsangeethakumari.g@gmail.com\nD83YNKBVRA\nThis file is meant for personal use by sangeethakumari.g@gmail.com only.\nSharing or publishing the contents in part or full is liable for legal action.')]]
```

```
# Observation : This code converts user query into a numerical embedding using the
# and compares this query embedding to all the document embeddings stored in the vectorstore
# and then finds the k chunks that are similar to the user query embeddings.
```

Retriever

```
retriever = vectorstore.as_retriever(
    search_type='similarity',
    search_kwargs={'k': 2}
)
```

```
rel_docs = retriever.get_relevant_documents("What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it ?")
```

```
/tmp/ipykernel_8942/861578063.py:1: LangChainDeprecationWarning: The method `BaseRetriever.get_relevant_documents` was deprecated in langchain-core 0.1.46 and will be removed in 1.0. Use `retriever.invoke` instead.
```

```
rel_docs = retriever.get_relevant_documents("What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it ?")
```

```
[Document(id='7b0f69cc-5274-4e5d-b036-72959bfd5de2', metadata={'title': 'The Merck Manual of Diagnosis & Therapy, 19th Edition', 'creator': 'Atop CHM to PDF Converter', 'total_pages': 4114, 'keywords': '', 'moddate': '2026-03-18T21:26:23+00:00', 'format': 'PDF 1.7', 'producer': 'pdf-lib (https://github.com/Hopding/pdf-lib)', 'creationdate': '2012-06-15T05:44:40+00:00', 'trapped': '', 'subject': '', 'file_path': '/content/medical_diagnosis_manual.pdf', 'creationDate': 'D:20120615054440Z', 'source': '/content/medical_diagnosis_manual.pdf', 'author': '', 'page': 172, 'modDate': 'D:20260318212623Z'}, page_content="Appendicitis\nAppendicitis is acute inflammation of the vermiform appendix, typically resulting in abdominal\npain, anorexia, and abdominal tenderness. Diagnosis is clinical, often supplemented by CT or\nultrasound. Treatment is surgical removal.\nIn the US, acute appendicitis is the most common cause of acute abdominal pain requiring surgery. Over\n5% of the population develops appendicitis at some point. It most commonly occurs in the teens and 20s\nbut may occur at any age.\nOther conditions affecting the appendix include carcinoids, cancer, villous adenomas, and diverticula. The\nappendix may also be affected by Crohn's disease or ulcerative colitis with pancolitis.\nThe Merck Manual of Diagnosis & Therapy, 19th Edition\nChapter 11. Acute Abdomen & Surgical Gastroenterology\n163\nsangeethakumari.g@gmail.com\nD83YNKBVRA\nThis file is meant for personal use by sangeethakumari.g@gmail.com only.\nSharing or publishing the contents in part or full is liable for legal action."),
```

```
Document(id='75eb5fd3-83ca-44c4-91c9-1840d02da6fc', metadata={'keywords': '', 'title': 'The Merck Manual of Diagnosis & Therapy, 19th Edition', 'producer': 'pdf-lib (https://github.com/Hopding/pdf-lib)', 'creationDate': 'D:20120615054440Z', 'trapped': '', 'total_pages': 4114, 'page': 173, 'file_path': '/content/medical_diagnosis_manual.pdf', 'format': 'PDF 1.7', 'author': '', 'moddate': '2026-03-18T21:26:23+00:00', 'modDate': 'D:20260318212623Z', 'creationdate': '2012-06-15T05:44:40+00:00', 'subject': '', 'creator': 'Atop CHM to PDF Converter', 'source': '/content/medical_diagnosis_manual.pdf'}, page_content="Etiology\nAppendicitis is thought to result from obstruction of the appendiceal lumen, typically by lymphoid\nhyperplasia, but occasionally by a fecalith, foreign body, or even worms. The obstruction leads to\n distention, bacterial overgrowth, ischemia, and inflammation. If untreated, necrosis, gangrene, and\nperforation occur. If the perforation is contained by the omentum, an appendiceal abscess results.\nSymptoms and Signs\nThe classic symptoms of acute appendicitis are epigastric or periumbilical pain followed by brief nausea,\nvomiting, and anorexia; after a few hours, the pain shifts to the right lower quadrant. Pain increases with\ncough and motion. Classic signs are right lower quadrant direct and rebound tenderness located at\nMcBurney's point (junction of the middle and outer thirds of the line joining the umbilicus to the anterior\nsuperior spine). Additional signs are pain felt in the right lower quadrant with palpation of the left lower\nquadrant (Rovsing sign), an increase in pain from passive extension of the right hip joint that stretches\nthe iliopsoas muscle (psoas sign), or pain caused by passive internal rotation of the flexed thigh\n(obturator sign). Low-grade fever (rectal temperature 37.7 to 38.3° C [100 to 101° F]) is common.\nUnfortunately, these classic findings appear in < 50% of patients. Many variations of symptoms and signs\noccur. Pain may not be localized, particularly in infants and children. Tenderness may be diffuse or, in rare\ninstances, absent. Bowel movements are usually less frequent or absent; if diarrhea is a sign, a\nretrocecal appendix should be suspected. RBCs or WBCs may be present in the urine. Atypical symptoms\nare common among elderly patients and pregnant women; in particular, pain is less severe and local\ntenderness is less marked.\nDiagnosis\n• Clinical evaluation\n• Abdominal CT if necessary\n• Ultrasound an option to CT\nWhen classic symptoms and signs are present, the diagnosis is clinical. In such patients, delaying\nlaparotomy to do imaging tests only increases the likelihood of perforation and subsequent complications.\nIn patients with atypical or equivocal findings, imaging studies should be done without delay. Contrast-")]
```

```
#Observation : We can observe that the two relevant chunks contain the answer to the
# If we increase the k value, there is a chance that we might find the answer in even
# This is a hyperparameter that we need to tune to get the best similarity from the
```

```
model_output = llm(
    "What are the common symptoms for appendicitis, and can it be cured via medic
    max_tokens=1000,
    temperature=2,
)
```

Llama.generate: prefix-match hit

```
model_output['choices'][0]['text']
```

'\n\nAppendicitis is a medical condition characterized by inflammation of the appendix, a small pouch connected to the large intestine in the lower right side of the abdomen. Some common symptoms of appendicitis include:\n\n1. Abdominal pain: This is typically localized in the lower right abdomen, starting as a mild cramp-like sensation and progressing into a sharp, persistent pain. The pain may worsen with movement or pressure on the abdomen.\n2. Loss of appetite and feeling sick: Nausea and vomiting are common symptoms of appendicitis. Some people may also experience a loss of interest in food and a general feeling of malaise or illness.\n3. Fever: A fever of 101°F (38.3°C) or higher is common with appendicitis. The body produces a fever as a way to help fight the infection causing the inflammation.\n4. Constipation or diarrhea: While some people experience constipation, others may have diarrhea as a result of the condition. This can depend on how badly the appendix is inflamed and whether it has ruptured.\n5. Rebound tenderness: When a doctor gently presses on the abdomen and then releases the pressure, there may be discomfort or pain in the area where the appendix is located, indicating rebound tenderness.\n\nIf left untreated, appendicitis can result in rupture of the appendix, which can lead to serious complications, including peritonitis (inflammation of the abdominal cavity), sepsis (a potentially life-threatening condition caused by infection), and death. Therefore, prompt medical attention is necessary if you suspect that you have appendicitis.\n\nThe only cure for appendicitis is surgery to remove the infected appendix, called an appendectomy. This can be performed using laparoscopic or open techniques. Laparoscopic appendectomy involves making small incisions and inserting a camera and instruments to remove the appendix through one of these incisions. Open appendectomy involves making a larger incision to directly access and remove the appendix. Both procedures have their benefits and risks, and your healthcare provider will discuss which option is best for you based on your individual circumstances.\n\nIn summary, if you suspect that you may have appendicitis, seek medical attention immediately as it can lead to serious complications if left untreated. While there is no cure for appendicitis via medication, the only treatment option available is an appendectomy.'

#Observation : We can observe that the result is limited to the 1000 token for the

System and User Prompt Template

```
qna_system_message = """
You are an medical assistant whose work is to review the report and provide the app
User input will have the context required by you to answer user questions.
This context will begin with the token: ###Context.
The context contains references to specific portions of a document relevant to the

User questions will begin with the token: ###Question.

Please answer only using the context provided in the input. Do not mention anything

If the answer is not found in the context, respond "I don't know".
"""
```



```

qna_user_message_template = """
###Context
Here are some documents that are relevant to the question mentioned below.
{context}

###Question
{question}
"""

```

Response Function

```

def generate_rag_response(user_input,k=3,max_tokens=128,temperature=0,top_p=0.95,tc
    global qna_system_message,qna_user_message_template
    # Retrieve relevant document chunks
    relevant_document_chunks = retriever.get_relevant_documents(query=user_input,k=
    context_list = [d.page_content for d in relevant_document_chunks]

    # Combine document chunks into a single context
    context_for_query = ". ".join(context_list)

    user_message = qna_user_message_template.replace('{context}', context_for_query)
    user_message = user_message.replace('{question}', user_input)

    prompt = qna_system_message + '\n' + user_message

    # Generate the response
    try:
        response = llm(
            prompt=prompt,
            max_tokens=max_tokens,
            temperature=temperature,
            top_p=top_p,
            top_k=top_k
        )

        # Extract and print the model's response
        response = response['choices'][0]['text'].strip()
    except Exception as e:
        response = f'Sorry, I encountered the following error: \n {e}'

    return response

```

Question Answering using RAG

Query 1: What is the protocol for managing sepsis in a critical care unit?

```

user_input_1 = "What is the protocol for managing sepsis in a critical care unit?"
generate_rag_response(user_input_1,top_k=20)

```

Llama.generate: prefix-match hit

"Based on the context provided, the protocol for managing sepsis in a critical care unit includes:\n1. Monitoring: The patient's vital signs, including ECG, systolic, diastolic, and mean BP, respiratory rate and depth, pulse oximetry, urine flow by indwelling bladder catheter, body temperature, and clinical status are closely monitored.\n2. Diagnosis: If sepsis or septic shock is suspected, cultures are obtained of blood and any other appropriate specimens.\n3. Treatment:\n a."

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
user_input_2 = "What are the common symptoms for appendicitis, and can it be cured  
generate_rag_response(user_input_2,top_k=20)
```

Llama.generate: prefix-match hit

"Based on the context provided, the common symptoms for appendicitis include epigastric or periumbilical pain followed by brief nausea, vomiting, and anorexia; after a few hours, the pain shifts to the right lower quadrant. Pain increases with cough and motion. Classic signs are right lower quadrant direct and rebound tenderness located at McBurney's point (junction of the middle and outer thirds of the line joining the umbilicus to the anterior superior spine). Additional signs include pain felt in the right lower quadrant with palpation of the left"

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
user_input_3 = "What are the effective treatments or solutions for addressing sudden  
generate_rag_response(user_input_3,top_k=20)
```

Llama.generate: prefix-match hit

'Based on the context provided, the condition being described is Alopecia Areata. The treatment options mentioned include daily hair counts to quantify hair loss, microscopic hair examination, scalp biopsy, and immunofluorescence studies. These tests are used for definitive diagnosis and to differentiate scarring from non-scarring forms of alopecia. Treatment options for Alopecia Areata include topical Minoxidil (2% for women, 2% or 5% for men) which prolongs the anagen growth phase and enlarges miniaturized'

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
user_input_4 = "What treatments are recommended for a person who has sustained a physical  
generate_rag_response(user_input_4,top_k=20)
```

Llama.generate: prefix-match hit

'I. Supportive care including preventing systemic complications due to immobilization, providing good nutrition, and preventing pressure ulcers is recommended.\nII. Physical therapy to prevent limb contractures is also suggested for some patients.\nIII. No specific treatment exists for the injury itself, but support groups may provide assistance to families of brain-injured patients.\nIV. A prolonged period of rehabilitation, particularly in cognitive and emotional areas, is often required for patients whose coma exceeds 24 hours and have major persistent neurologic sequelae.'

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
user_input_5 = "What are the necessary precautions and treatment steps for a person  
generate_rag_response(user_input_5,top_k=20)
```

Llama.generate: prefix-match hit

"Based on the context provided, the following are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip:\n1. The first priority is to ensure that the person's airway is clear and they are breathing properly. If there are signs of shock or severe bleeding, call for emergency medical help immediately.\n2. Splinting the leg to prevent further injury and decrease pain is necessary. This can be done using a nonrigid or noncircumferential device such as a splint.\n3. Pain should be treated with opioids, if"

```
#Observation : The answer is clear, concise, and focused, without any unnecessary i  
# All the queries are answered from the context provided – the medical manual docum
```

Query 6: What are the necessary steps to join a College ?

```
user_input_6 = "What are the necessary steps to join a College ?"  
generate_rag_response(user_input_6,top_k=20)
```

Llama.generate: prefix-match hit

"I don't know. The context provided does not contain information about joining a college."

```
# Observation : If we look at the system prompt, we explicitly mentioned that the q  
# As expected, the model has done its job well. It has eliminated hallucination.
```

Question Answering using RAG - Changing Paramaters - Changing the K value to 1

Query 1_1: What is the protocol for managing sepsis in a critical care unit?

```
user_input_1 = "What is the protocol for managing sepsis in a critical care unit?"  
generate_rag_response(user_input_1,top_k=1)
```

Llama.generate: prefix-match hit

"Based on the context provided, the protocol for managing sepsis in a critical care unit includes:\n1. Monitoring: The patient's vital signs, including ECG, systolic, diastolic, and mean BP, respiratory rate and depth, pulse oximetry, urine flow by indwelling bladder catheter, body temperature, and clinical status are closely monitored.\n2. Diagnosis: If sepsis or septic shock is suspected, cultures are obtained of blood and any other appropriate specimens.\n3. Treatment:\n a."

```
user_input_1 = "What is the protocol for managing sepsis in a critical care unit?"  
generate_rag_response(user_input_1,top_k=1, max_tokens=300)
```

Llama.generate: prefix-match hit

"Based on the context provided, the protocol for managing sepsis in a critical care unit includes:\n1. Monitoring: The patient's vital signs, including ECG, systolic, diastolic, and mean BP, respiratory rate and depth, pulse oximetry, urine flow by indwelling bladder catheter, body temperature, and clinical status are closely monitored.\n2. Diagnosis: If sepsis or septic shock is suspected, cultures are obtained of blood and any other appropriate specimens.\n3. Treatment:\n a. Antibiotics: Empiric antibiotics are given after appropriate cultures are obtained. Early treatment with an appropriate antimicrobial regimen appears to improve survival. Continuing therapy involves adjusting antibiotics according to the results of culture and susceptibility testing, surgically draining any abscesses, and removing any internal devices that are the suspected source of bacteria.\n b. Fluid resuscitation: In major hemorrhage or in patients with signs of high right-sided pressure or acute MI, 1 L (or 20 mL/kg in children) of 0.9% saline is infused over 15 minutes. Ringer's lactate may be used instead. Further fluid therapy is based on the underlying condition and may require monitoring of CVP"

#observation : By increasing the max_token limit to 300, the model has enough "room" to successfully move past the initial diagnosis and into detailed Antibiotic and

Query 2_1: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
user_input_2 = "What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?"
generate_rag_response(user_input_2, top_k=1)
```

Llama.generate: prefix-match hit

"Based on the context provided, the common symptoms for appendicitis include epigastric or periumbilical pain followed by brief nausea, vomiting, and anorexia; after a few hours, the pain shifts to the right lower quadrant. Pain increases with cough and motion. Classic signs are right lower quadrant direct and rebound tenderness located at McBurney's point (junction of the middle and outer thirds of the line joining the umbilicus to the anterior superior spine). Additional signs include pain felt in the right lower quadrant with palpation of the left"

```
user_input_2 = "What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?"
generate_rag_response(user_input_2, top_k=5)
```

Llama.generate: prefix-match hit

"Based on the context provided, the common symptoms for appendicitis include epigastric or periumbilical pain followed by brief nausea, vomiting, and anorexia; after a few hours, the pain shifts to the right lower quadrant. Pain increases with cough and motion. Classic signs are right lower quadrant direct and rebound tenderness located at McBurney's point (junction of the middle and outer thirds of the line joining the umbilicus to the anterior superior spine). Additional signs include pain felt in the right lower quadrant with palpation of the left"

Observation : Changing the top_k value from 1 to 5 did not change the response. The model provides an identical, high-quality clinical description.

Query 3_1: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
user_input_3 = "What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?"
generate_rag_response(user_input_3, top_k=1)
```

Llama.generate: prefix-match hit

'Based on the context provided, the condition being described is Alopecia Areata. The treatment options mentioned include daily hair counts to quantify hair loss, microscopic examination of hairs, scalp biopsy, and immunofluorescence studies to help identify underlying disorders. The possible causes are thought to be autoimmune in nature and may be triggered by unclear environmental factors. Treatment options for Alopecia Areata include minoxidil (2% for women, 2% or 5% for men) which prolongs the anagen growth phase and enlarges mini'

```
user_input_3 = "What are the effective treatments or solutions for addressing sudden hair loss?"
generate_rag_response(user_input_3,top_k=1,top_p=0.5,temperature=0.7)
```

Llama.generate: prefix-match hit

'Based on the context provided, the condition being described is Alopecia Areata. The treatment options mentioned include daily hair counts to quantify hair loss, microscopic hair examination, scalp biopsy, and immunofluorescence studies for diagnosis. For treatment, minoxidil (2% for women, 2% or 5% for men) is effective in prolonging the anagen growth phase and enlarging miniaturized follicles into mature terminal hairs. Finasteride, which inhibits the 5 α -reductase enzyme, can also be'

*#Observation : The introduction of temperature=0.7 and top_p=0.5 allows the model to generate more detailed responses.
The core response remains the same .*

Query 4_1: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
user_input_4 = "What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?"
generate_rag_response(user_input_4,top_k=1)
```

Llama.generate: prefix-match hit

'I. Supportive care including preventing systemic complications due to immobilization, providing good nutrition, and preventing pressure ulcers is recommended.\nII. Physical therapy to prevent limb contractures is also suggested for some patients.\nIII. No specific treatment exists for the injury itself, but support groups may provide assistance to families of brain-injured patients.\nIV. A prolonged period of rehabilitation, particularly in cognitive and emotional areas, is often required for patients whose coma exceeds 24 hours and have major persistent neurologic sequelae.'

```
user_input_4 = "What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?"
generate_rag_response(user_input_4,top_k=15)
```

Llama.generate: prefix-match hit

'I. Supportive care including preventing systemic complications due to immobilization, providing good nutrition, and preventing pressure ulcers is recommended.\nII. Physical therapy to prevent limb contractures is also suggested for some patients.\nIII. No specific treatment exists for the injury itself, but support groups may provide assistance to families of brain-injured patients.\nIV. A prolonged period of rehabilitation, particularly in cognitive and emotional areas, is often required for patients whose coma exceeds 24 hours and have major persistent neurologic sequelae.'

#Observation : Changing the top_k value from 1 to 15 , the response did not change

Query 5_1: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
user_input_5 = "What are the necessary precautions and treatment steps for a person  
generate_rag_response(user_input_5,top_k=1)
```

Llama.generate: prefix-match hit

"Based on the context provided, the following are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip:\n1. The first priority is to ensure that the person's airway is clear and they are breathing properly. If there are signs of shock or severe bleeding, call for emergency medical help immediately.\n2. Splinting the leg to prevent further injury and decrease pain is necessary. This can be done using a nonrigid or noncircumferential device such as a splint.\n3. Pain should be treated with opioids, if"

```
user_input_5 = "What are the necessary precautions and treatment steps for a person  
generate_rag_response(user_input_5,top_k=1,max_tokens=300)
```

Llama.generate: prefix-match hit

"Based on the context provided, the following are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip:\n1. The first priority is to ensure that the person's airway is clear and they are breathing properly. If there are signs of shock or severe bleeding, call for emergency medical help immediately.\n2. Splinting the leg to prevent further injury and decrease pain is necessary. This can be done using a nonrigid or noncircumferential device such as a splint.\n3. Pain should be treated with opioids, if necessary.\n4. Definitive treatment may involve reduction, which requires analgesia or sedation. Closed reduction without skin incision is preferred when possible, while open reduction with skin incision is done when necessary.\n5. RICE (rest, ice, compression, and elevation) should be applied to the injured leg to prevent further injury and speed healing. Ice should be applied intermittently during the first 24 to 48 hours for 15 to 20 minutes as often as possible. The injury can be compressed using a splint, an elastic bandage, or a Jones compression dressing.\n6. For patients with a lower-limb prosthesis, maintaining body alignment is important to prevent skin breakdown. This may involve switching to a thicker or thinner liner or sock,"

*#Observation : By increaing the max tokens. The response is more detailed yet the
#Maybe increaing the max token to 500 may result in complete answer.*

Query 6_1: What are the necessary steps to join a College ?

```
user_input_6 = "What are the necessary steps to join a College ?"  
generate_rag_response(user_input_6,top_k=20)
```

Llama.generate: prefix-match hit

"I don't know. The context provided does not contain information about joining a college."

*# Observation : If we look at the system prompt, we explicitly mentioned that the q
As expected, the model has done its job well. It has eliminated hallucination.*

Fine-tuning

Query 1: What is the protocol for managing sepsis in a critical care unit?

```
user_input_1 = "What is the protocol for managing sepsis in a critical care unit?"  
generate_rag_response(user_input_1,max_tokens=100)
```

Llama.generate: prefix-match hit

"Based on the context provided, the protocol for managing sepsis in a critical care unit includes:\n1. Monitoring: The patient's vital signs, including ECG, systolic, diastolic, and mean BP, respiratory rate and depth, pulse oximetry, urine flow by indwelling bladder catheter, body temperature, and clinical status are closely monitored.\n2. Diagnosis: If sepsis or septic shock is suspected, cultures are obtained of blood and any other appropriate specimens.\n3. Treatment:\n a. Antibiotics: Empiric antibiotics are given after appropriate cultures are obtained. Early treatment with an appropriate antimicrobial regimen appears to improve survival. Continuing therapy involves adjusting antibiotics according to the results of culture and susceptibility testing, surgically draining any abscesses, and removing any internal devices that"

```
user_input_2 = "What is the protocol for managing sepsis in a critical care unit?"  
generate_rag_response(user_input_2,max_tokens=200 , temperature=0.1)
```

Llama.generate: prefix-match hit

"Based on the context provided, the protocol for managing sepsis in a critical care unit includes:\n1. Monitoring: The patient's vital signs, including ECG, systolic, diastolic, and mean BP, respiratory rate and depth, pulse oximetry, urine flow by indwelling bladder catheter, body temperature, and clinical status are closely monitored.\n2. Diagnosis: If sepsis or septic shock is suspected, cultures are obtained of blood and any other appropriate specimens.\n3. Treatment:\n a. Antibiotics: Empiric antibiotics are given after appropriate cultures are obtained. Early treatment with an appropriate antimicrobial regimen appears to improve survival. Continuing therapy involves adjusting antibiotics according to the results of culture and susceptibility testing, surgically draining any abscesses, and removing any internal devices that"

#Observation : Even if the max_tokens is increased by 100, the model still didn't generate the full answer.

#Even though the answer is not complete as the temperature is closer to 0, making the model more deterministic and focused on the context provided.

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
user_input_3 = "What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?"  
generate_rag_response(user_input_3,max_tokens=200 , top_k=20, temperature=0.1)
```

Llama.generate: prefix-match hit

"Based on the context provided, the condition being described is Alopecia Areata. The treatment options mentioned include daily hair counts to quantify hair loss, microscopic examination of hairs, scalp biopsy, and immunofluorescence studies. The possible causes are thought to be autoimmune in nature and may involve genetic susceptibility and environmental triggers. There is no definitive cure for Alopecia Areata, but treatments such as topical Minoxidil or Finasteride can help slow down hair loss and promote regrowth in some cases. However, it's important to note that these treatments are not effective for all causes of hair loss and may have side effects. If the condition persists and diagnosis is in doubt, a scalp biopsy may be necessary to differentiate scarring from non-scarring forms."

#Observation : Previously the answer was limited, as the query could be answered with more context.

#Now with the temperature is set to 0.1, making the model more deterministic and focused on the context provided.

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of

brain function?

```
user_input_4 = "What treatments are recommended for a person who has sustained a ph  
generate_rag_response(user_input_4,max_tokens=200 , top_k=20, temperature=0.1)
```

Llama.generate: prefix-match hit

'I. Supportive care including preventing systemic complications due to immobilization, providing good nutrition, and preventing pressure ulcers is recommended.\nII. Physical therapy to prevent limb contractures is also suggested for some patients.\nIII. No specific treatment exists for the injury itself, but support groups may provide assistance to families of brain-injured patients.\nIV. A prolonged period of rehabilitation, particularly in cognitive and emotional areas, is often required for patients whose coma exceeds 24 hours and have major persistent neurologic sequelae.'

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
user_input_5 = "What are the necessary precautions and treatment steps for a person  
generate_rag_response(user_input_5,max_tokens=200 , top_k=20, temperature=0.1) #Con
```

Llama.generate: prefix-match hit

"Based on the context provided, here's the answer:\n\nThe person with a fractured leg should first receive appropriate medical attention in an emergency department. The injury may require immobilization to prevent further damage to soft tissues and decrease pain. Splinting is usually used for this purpose, but definitive treatment such as reduction may also be necessary. Pain is typically treated with opioids. RICE (rest, ice, compression, elevation) can benefit the person by preventing further injury and speeding healing. For patients with a lower-limb prosthesis, maintaining body alignment is important to prevent skin breakdown. Rehabilitation should begin as soon as possible after surgery to increase strength and prevent atrophy on the unaffected side. Gradual mobilization of the affected limb usually results in full ambulation. The speed of rehabilitation depends on the type of surgery done. Patients are taught to do daily exercises to strengthen the trunk muscles and"

Fine-tuning

Fine-tuning - Changing parameters - increasing the temperature value to 0.9

Query 1: What is the protocol for managing sepsis in a critical care unit?

```
user_input_1 = "What is the protocol for managing sepsis in a critical care unit?"  
generate_rag_response(user_input_1,temperature=0.9 )
```

Llama.generate: prefix-match hit

'Based on the context, the protocol for managing sepsis in a critical care unit includes:\n1. Suspecting and diagnosing sepsis or septic shock based on symptoms such as shaking chills, persistent fever, altered sensorium, hypotension, and GI symptoms (abdominal pain, nausea, vomiting, diarrhea).\n2. Obtaining cultures of blood and any other appropriate specimens for diagnosis.\n3. Treat'


```
user_input_2 = "What is the protocol for managing sepsis in a critical care unit?"
generate_rag_response(user_input_2, temperature=0.9)
```

Llama.generate: prefix-match hit

'Based on the context provided, the management protocol for sepsis in a critical care unit includes:\n- Monitoring: ECG, systolic, diastolic, and mean BP preferably by intra-arterial catheter; respiratory rate and depth; pulse oximetry; urine flow by indwelling bladder catheter; body temperature; and clinical status, including sensorium (Glasgow Coma Scale).\n- Diagnosis: If bacteremia, sepsis, or septic shock is suspected, cultures are obtained of blood and any other appropriate specimens.\n- Treatment:\n - Antibiotics: Empiric antibiotics are given after appropriate cultures are obtained. Early treatment with an appropriate antimicrobial regimen appears to improve survival. Continuing therapy involves adjusting antibiotics according to the results of culture and susceptibility testing, surgically draining'

#Observation : Even if the max_tokens not changes, the model still didn't generate

#One of the reasons could be that the temperature is set to 0.9, making the model n

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
user_input_3 = "What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?"
generate_rag_response(user_input_3, top_k=20, temperature=.9)
```

Llama.generate: prefix-match hit

'Based on the context provided, the condition described in the question is likely Alopecia Areata. The effective treatments mentioned for this condition include daily hair counts to quantify hair loss, scalp biopsy to differentiate scarring from nonscarring forms, and treatment with minoxidil or finasteride. The possible causes behind sudden patchy hair loss include autoimmune disorders and exposure to unclear environmental triggers.'

#Observation : Now with the temperature is set .9 makes it a little more creative.

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
user_input_4 = "What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?"
generate_rag_response(user_input_4, top_k=20, temperature=.9)
```

Llama.generate: prefix-match hit

'You should answer: The recommended treatments for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function include preventive measures to prevent systemic complications such as pneumonia, UTI, and thromboembolic disease. Providing good nutrition and preventing pressure ulcers are also important. Some patients may require physical therapy to prevent limb contractures. Rehabilitation services should be planned early for those whose coma exceeds 24 hours.'

#Observation : Now with the temperature is set .9 makes it a little more creative.

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered

for their care and recovery?

```
user_input_5 = "What are the necessary precautions and treatment steps for a person  
generate_rag_response(user_input_5,max_tokens=200 , top_k=20, temperature=.9) #Comp
```

Llama.generate: prefix-match hit

"From the context, the treatment for a fractured leg includes: splinting to prevent further injury, pain management with opioids, definitive treatment (reduction), immobilization with casting or surgical hardware, RICE therapy, and rehabilitation as soon as possible after surgery. The person should also be evaluated for signs of ischemia, such as absent pulses, marked pallor, coolness distal to the injury, and severe pain. It's important for them to maintain body alignment if they have a lower-limb prosthesis and seek prompt medical attention if there are signs of skin breakdown. The speed of rehabilitation depends on the type of surgery done. Additionally, closed injuries that do not involve blood vessels or nerves are least likely to result in serious complications."

#Observation : Now with the temperature is set .9 makes it a little more creative.

Output Evaluation

Let us now use the LLM-as-a-judge method to check the quality of the RAG system on two parameters - retrieval and generation. We illustrate this evaluation based on the answers generated to the question from the previous section.

- We are using the same Mistral model for evaluation, so basically here the LLM is rating itself on how well he has performed in the task.

```
groundedness_rater_system_message = """  
You are tasked with rating AI generated answers to questions posed by users.  
You will be presented a question, context used by the AI system to generate the ans  
In the input, the question will begin with ###Question, the context will begin with  
  
Evaluation criteria:  
The task is to judge the extent to which the metric is followed by the answer.  
1 - The metric is not followed at all  
2 - The metric is followed only to a limited extent  
3 - The metric is followed to a good extent  
4 - The metric is followed mostly  
5 - The metric is followed completely  
  
Metric:  
The answer should be derived only from the information presented in the context  
  
Instructions:  
1. First write down the steps that are needed to evaluate the answer as per the met  
2. Give a step-by-step explanation if the answer adheres to the metric considering  
3. Next, evaluate the extent to which the metric is followed.  
4. Use the previous information to rate the answer using the evaluation criteria and  
"""
```

```
relevance_rater_system_message = """You are tasked with rating AI generated answers  
You will be presented a question, context used by the AI system to generate the ans  
In the input, the question will begin with ###Question, the context will begin with
```

Evaluation criteria:

The task is to judge the extent to which the metric is followed by the answer.

- 1 - The metric is not followed at all
- 2 - The metric is followed only to a limited extent
- 3 - The metric is followed to a good extent
- 4 - The metric is followed mostly
- 5 - The metric is followed completely

Metric:

Relevance measures how well the answer addresses the main aspects of the question, Consider whether all and only the important aspects are contained in the answer when

Instructions:

- 1. First write down the steps that are needed to evaluate the context as per the metric
- 2. Give a step-by-step explanation if the context adheres to the metric considering
- 3. Next, evaluate the extent to which the metric is followed.
- 4. Use the previous information to rate the context using the evaluation criteria and

```
user_message_template = """
###Question
{question}

###Context
{context}

###Answer
{answer}
"""
```

```
def generate_ground_relevance_response(user_input,k=3,max_tokens=128,temperature=0,
    global qna_system_message,qna_user_message_template
    # Retrieve relevant document chunks
    relevant_document_chunks = retriever.get_relevant_documents(query=user_input,k=
    context_list = [d.page_content for d in relevant_document_chunks]
    context_for_query = ". ".join(context_list)

    # Combine user_prompt and system_message to create the prompt
    prompt = f"""[INST]{qna_system_message}\n
        {'user': {qna_user_message_template.format(context=context_for_query,
        [/INST]}"""

    response = llm(
        prompt=prompt,
        max_tokens=max_tokens,
        temperature=temperature,
        top_p=top_p,
        top_k=top_k,
        stop=['INST'],
    )

    answer = response["choices"][0]["text"]

    # Combine user_prompt and system_message to create the prompt
    groundedness_prompt = f"""[INST]{groundedness_rater_system_message}\n
        {'user': {user_message_template.format(context=context_for_query,
        [/INST]}"""

    # Combine user_prompt and system_message to create the prompt
```

```

relevance_prompt = f"""[INST]{relevance_rater_system_message}\n
    {'user': {user_message_template.format(context=context_for_query,
[/INST]}"""

response_1 = llm(
    prompt=groundedness_prompt,
    max_tokens=max_tokens,
    temperature=temperature,
    top_p=top_p,
    top_k=top_k,
    stop=['INST'],
)

response_2 = llm(
    prompt=relevance_prompt,
    max_tokens=max_tokens,
    temperature=temperature,
    top_p=top_p,
    top_k=top_k,
    stop=['INST'],
)

return response_1['choices'][0]['text'], response_2['choices'][0]['text']

```

Query 1: What is the protocol for managing sepsis in a critical care unit?

```

ground, rel = generate_ground_relevance_response(user_input="What is the protocol for
print(ground, end="\n\n")
print("_____")
print(rel)

```

```

Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
Llama.generate: prefix-match hit

```

Steps to evaluate the answer:

1. Identify the key information related to managing sepsis in a critical care unit from the context.
2. Compare the AI generated answer with the identified key information.
3. Determine if the answer is derived only from the information presented in the context.

Explanation:

The AI generated answer mentions monitoring for signs of sepsis or septic shock, obtaining cultures if suspected, and administering antibiotics after appropriate cultures are obtained. The context discusses monitoring ICU patients for various conditions including sepsis or septic shock, obtaining blood tests to help detect problems early, and treating shock with fluid resuscitation and mechanical ventilation.

The AI generated answer adheres to the metric as it mentions the specific signs of sepsis or septic shock to monitor for, obtaining cultures if suspected, and administering antibiotics after appropriate cultures are obtained, all of which is derived from the context.

Rating:

Based on the evaluation criteria, I would rate the answer as a 5 since it follows the metric completely by deriving the answer only from the information presented in the context.

Steps to evaluate the context as per the relevance metric:

1. Identify the main aspects of the question: managing sepsis in a critical care unit.
2. Determine if the context provides information on these main aspects.
3. Check if all important aspects are contained in the answer and if any irrelevant information is excluded.

The context adheres to the relevance metric considering the question as follows:

1. The context discusses sepsis and septic shock, which are relevant to managing sepsis in a critical care unit.
2. It mentions monitoring for signs of sepsis or septic shock, obtaining cultures, and administering antibiotics, all of which are important aspects of managing sepsis in a critical care unit.
3. The context also includes some irrelevant information about the ICU setting, but it does not significantly detract from the relevance of the answer to the question.

The extent to which the metric is followed:

The context follows the relevance metric mostly as it provides comprehensive information on managing sepsis in a critical care unit and covers all important aspects mentioned in the question. However, some irrelevant information is included but does not significantly impact the overall relevance of the answer.

Rating the context using the evaluation criteria:

Based on the above analysis, I would rate the context as a 4 (The metric is followed mostly) according to the evaluation criteria.

#Observation : Groundedness Evaluation – scored as 5. The judge found that the AI-

*#Observation : Relevance Evaluation – scored as 4. The judge noted that the context
#however one answer is irrelevant about the ICU setting. Hence the score 4*

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
ground,rel = generate_ground_relevance_response(user_input="What are the common sym  
print(ground,end="\n\n")  
print("_____  
print(rel)
```

Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
Llama.generate: prefix-match hit

Steps to evaluate the answer:

1. Identify the information in the context related to common symptoms and treatment for appendicitis.
2. Check if the AI generated answer includes only the information from the context.
3. Evaluate the extent to which the metric is followed.

Explanation:

The AI generated answer includes all the common symptoms for appendicitis mentioned in the context, such as epigastric or periumbilical pain, nausea, vomiting, anorexia, shifting pain to the right lower quadrant, increase in pain with cough and motion, direct and rebound tenderness at McBurney's point, Rovsing sign, psoas sign, obturator sign, and low-grade fever. The answer also mentions that appendicitis cannot be cured via medicine alone and the standard treatment is surgical removal of the appendix, which is consistent with the information in the context.

Therefore, the metric is followed completely as the answer is derived only from the information presented in the context.

Rating:

Based on the evaluation criteria, I would rate the answer a 5 for following the metric completely.

Steps to evaluate the context as per the relevance metric:

1. Identify the main aspects of the question: common symptoms for appendicitis and whether it can be cured via medicine or if surgery is required.
2. Determine if all important aspects are contained in the context: The context provides a detailed explanation of appendicitis, its causes, symptoms, diagnosis, and treatment. It mentions that the standard treatment for appendicitis is surgical removal. However, it does not explicitly state whether appendicitis can be cured via medicine or not.
3. Check if only irrelevant information is missing: The context provides a comprehensive explanation of appendicitis and its symptoms. It also discusses other conditions that can affect the appendix but does not provide information on whether appendicitis can be cured via medicine or not.

The context adheres to the metric to a good extent as it covers most of the important aspects of the question, but it lacks explicit information about whether appendicitis can be cured via medicine or not.

Rating: 3 (Followed to a good extent)

#Observation : Groundedness Evaluation – scored as 5. The judge found that the AI-

#Observation : Relevance Evaluation – scored as 3. While the context provided a det

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
ground, rel = generate_ground_relevance_response(user_input="What are the effective  
print(ground, end="\n\n")  
print(rel)
```

```
Llama.generate: prefix-match hit  
Llama.generate: prefix-match hit  
Llama.generate: prefix-match hit
```

Steps to evaluate the answer:

1. Identify the main question and the specific information being asked for in the question.
2. Read through the context provided to identify any relevant information related to the question.
3. Determine if the AI generated answer is derived solely from the information presented in the context.

Explanation:

The main question asks about effective treatments or solutions for addressing sudden patchy hair loss, also known as alopecia areata, and possible causes behind it. The context provided discusses various types of alopecia, including androgenetic alopecia and alopecia areata, their causes, and potential treatments.

The AI generated answer includes several treatment options for addressing sudden patchy hair loss (alopecia areata), which are all mentioned in the context. The answer also correctly identifies alopecia areata as an autoimmune disorder and mentions that it affects genetically susceptible people exposed to unclear environmental triggers, which is consistent with the information provided in the context.

Therefore, based on the given metric, the AI generated answer adheres completely to the requirement of being derived only from the information presented in the context.

Rating:

Based on the evaluation criteria and the complete adherence of the AI generated answer to the metric, I would rate the answer as a 5.

Steps to evaluate context as per the relevance metric:

1. Identify the main aspects of the question: effective treatments or solutions for addressing sudden patchy hair loss and possible causes behind it.
2. Determine if the context provides information on these main aspects.
3. Check if all important aspects are contained in the answer when evaluating relevance.

The context adheres to the metric considering the question as follows:

1. The context mentions several causes of hair loss, including alopecia areata, which is a type of sudden patchy hair loss.
2. The context provides information on treatments for alopecia areata, such as topical Minoxidil, Finasteride, corticosteroids, retinoids, or immunosuppressants.
3. The context also mentions the importance of scalp biopsy for definitive diagnosis and other diagnostic tests like fungal and bacterial cultures and immunofluorescence studies.
4. Therefore, the context covers all important aspects of the question regarding possible causes and treatments for sudden patchy hair loss.

The metric is followed mostly as the context provides comprehensive information on both the causes and treatments for sudden patchy hair loss.

Rating: 4 (The metric is followed mostly)

#Observation : Groundedness Evaluation – scored as 5. The judge found that the AI-

#Observation : Relevance Evaluation – scored as 4. While the context provided a det

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?


```
ground, rel = generate_ground_relevance_response(user_input="What treatments are recommended for a person with brain damage resulting in impairment of brain function?")
print(ground, end="\n\n")
print(rel)
```

Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
Llama.generate: prefix-match hit

Steps to evaluate the answer:

1. Identify the key information in the context related to treatments for brain damage.
2. Compare the information identified with the AI generated answer to ensure that the answer is derived only from the context and not any external sources.
3. Evaluate the extent to which the metric is followed.

Explanation:

The AI generated answer mentions all the treatments mentioned in the context, including supportive care and physical therapy to prevent limb contractures, as well as the recommendation for families to seek assistance from support groups and the need for rehabilitation services for prolonged periods. Therefore, the answer is derived entirely from the information presented in the context.

Evaluation:

The metric is followed completely. The AI generated answer includes all the treatments mentioned in the context and does not include any external information.

Rating:

Based on the evaluation criteria, I would rate the answer as a 5, indicating that the metric is followed completely.

Steps to evaluate the context as per the relevance metric:

1. Identify the main aspects of the question: The question asks about treatments recommended for a person with brain damage resulting in impairment of brain function.
2. Determine if all and only the important aspects are contained in the answer: The context mentions that there is no specific treatment for brain damage, but supportive care should be provided. Supportive care includes preventing systemic complications due to immobilization, providing good nutrition, preventing pressure ulcers, and in some cases, physical therapy to prevent limb contractures. There is also a recommendation for families to seek assistance from support groups and rehabilitation services for prolonged periods. All the important aspects of the question are addressed in the context.

The metric is followed completely as all the important aspects of the question are contained in the answer.

Rating: 5 (The metric is followed completely)

#Observation : Groundedness Evaluation – scored as 5. The judge found that the AI-

#Observation : Relevance Evaluation – scored as 5. The answer addressed all key asp

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
ground, rel = generate_ground_relevance_response(user_input="What are the necessary
```

```
print(ground,end="\n\n")
print(rel)
```

Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
Llama.generate: prefix-match hit

Steps to evaluate the answer:

1. Identify the information in the context related to precautions and treatment steps for a person who has fractured their leg during a hiking trip.
2. Determine if the AI generated answer includes only the information derived from the context.
3. Evaluate the extent to which the metric is followed.

Explanation:

The AI generated answer includes several points that are directly related to the context, such as immobilizing the injured leg, treating pain with opioids, performing reduction (with analgesia or sedation), using RICE, starting rehabilitation as soon as possible, and avoiding heavy lifting or prolonged sitting. These points can be traced back to the information provided in the context.

Therefore, the metric is followed mostly since the answer includes all the necessary precautions and treatment steps for a person who has fractured their leg, derived directly from the context.

Rating:

Based on the evaluation criteria, I would rate the AI generated answer as a 4 (The metric is followed mostly).

Steps to evaluate the context as per the relevance metric:

1. Identify the main aspects of the question: necessary precautions, treatment steps, care, and recovery for a person who has fractured their leg during a hiking trip.
2. Determine if all the important aspects are contained in the context: yes, the context covers necessary precautions (immobilization, pain management), treatment steps (splinting, casting, reduction, RICE), care and recovery (rehabilitation, exercises, avoiding heavy lifting).

The context adheres to the relevance metric as it addresses all the main aspects of the question.

Evaluation:

- 1 - The metric is not followed at all: This rating is not applicable as the context follows the relevance metric.
- 2 - The metric is followed only to a limited extent: This rating is also not applicable as the context follows the relevance metric completely.
- 3 - The metric is followed to a good extent: This rating could be given if some aspects of the question were only partially addressed in the context but most were covered.
- 4 - The metric is followed mostly: This rating could be given if one or two important aspects were missing from the context.
- 5 - The metric is followed completely: This rating is applicable as all the main aspects of the question are contained in the context.

Based on the evaluation, I would rate the context a 5 (The metric is followed completely).

#Observation : Groundedness Evaluation score is 4. The judge noted the AI's response and Identified the main aspects of the question: necessary precautions, treatment steps, care, and recovery for a person who has fractured their leg during a hiking trip. However the judge felt it is mostly derived rather than getting from the context.

#Relevance Evaluation score is 5 : The judge have found the answer has followed al

Actionable Insights and Business Recommendations

Following are Actionanble Insights **and** general recommendations:

Vector database creation time increases **as** the number of pages **in** the PDF grows.

Retrieval parameter k **is** very important, since answers can be spread across multiple

Chunk_overlap helps maintain continuity, especially when context spans across chunk

Max_tokens should depend on how complex the query **is** – complex queries give more de

Prompt design **and** temperature tuning help control both response length **and** creativi

RAG parameters should be continuously adjusted based on the specific use case **for** b

In evaluation, focus more on groundedness **and** relevance to ensure the output **is** rel

Business Recommendations:

The temparature should be set to lowest **for** medical assistant.

Using the LLM-**as-a-judge** method to ensure the AI output consistent **in** healthcare sy

Power Ahead
